

PART B

A medical clinic needs a database to manage doctors, patients, appointments, and treatments.

Requirements:

- Doctors have DoctorID, name, specialization, and contact number.
- Patients have PatientID, name, address, and phone number.
- A doctor can attend to many patients; a patient can be treated by multiple doctors.
- Appointments exist between doctors and patients. Each appointment has AppointmentID, date, time, and status.
- Each appointment may include multiple treatments. Treatments have TreatmentID, name, and type.
- Treatment details include dosage, instructions, and treatment duration.

- a) Construct an Entity Relationship Diagram (ERD) by using Crow's Foot notation, complete with all required entities, attributes, and relationships with their names, cardinalities, and connectivities as well as the primary and foreign keys that reflect all requirements and business rules that have been given by that company. (12.5 marks)